

Ph.D. Qualifying Exam in Probability
May 1989

1. Let X_n be independent identically distributed random variables and $S_n = \sum_{i=1}^n X_i$.

Suppose
$$E\left[\sup_n \left|\frac{S_n}{n}\right|\right] < \infty.$$

Show that

$$E[|X_1| \log^+ |X_1|] < \infty.$$

Where
$$\log^+ x = \begin{cases} \log x & \text{if } x \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Hint:
$$C_n = \prod_{j=1}^n P[|X_j| \leq j] \rightarrow c > 0.$$

If $T = \inf \{n: |X_n| > n\}$ then

$$\sum_1^\infty \frac{1}{n} \int_{(T=n)} |X_n| dp = \sum_1^\infty \frac{C_{n-1}}{n} \int_{|X_1| > n} |X_1| dp$$

$$\sum_1^\infty \frac{1}{n} \int_{(T=n)} |S_{n-1}| dp < \infty.$$

2. Let X_n be independent Poisson variables. Suppose $\sum_1^\infty X_n < \infty$ a.s.
Show that $\sum_1^\infty E[X_n] < \infty$.

Hint. Borel Cantelli.

3. a) Define the notion of conditional expectation. In particular explain the notion $E[X|Y]$ where X and Y are random variables.

b) Suppose X, Y are random variables such that $E(X^2), E(Y^2) < \infty$

and $E[X|Y] = Y$

$E[Y|X] = X$

Show that $X = Y$ a.s.

4. a) A sequence of characteristic functions on $\mathbb{R}^1$ converging pointwise to a characteristic function does so uniformly on compact sets. Explain why - no need to give complete proofs.

b) Let $b_n \in \mathbb{R}$ and f a characteristic function not identically equal to 1. Suppose $f(b_n t)$ converges to a characteristic function. Show that $\lim b_n = b \in \mathbb{R}$.

c) Let $\phi(t) = \frac{1}{8}[1 + 7e^{it}]$. Show that ϕ is a characteristic function but that $|\phi|$ is not.

Hint. $|\phi|^2$ is the characteristic function of a measure concentrated on $\{-1, 0, 1\}$.

If $|\phi|$ is the characteristic function of a measure m .

Then $|\phi|^2$ is the characteristic function of $m * m$.

5. Let X_t be a standard Brownian motion.

(i) Show that $(X_t), (X_t^2 - t)$ and $e^{uX_t - \frac{1}{2}u^2t}$ are all martingales.

(ii) Let $T_a(w) = \inf\{t: X_t(w) = a\}$ for $a > 0$. The inf being defined to be $+\infty$ if the set is empty. Using (1) Compute $E[e^{-\lambda T_a}]$ for $\lambda > 0$, and show that $T_a < \infty$ a.s. and $E[T_a] = \infty$.

6. State and sketch the proof of the martingale convergence theorem.

7. Using the martingale convergence theorem prove the following:

Let F_n be an increasing sequence of σ -fields, X_n a sequence of random variables dominated by an integrable random variable Y : $|X_n| \leq Y$. Suppose $X_n \rightarrow X$ a.s. Then $E[X_n|F_m] \rightarrow E[X|F_\infty]$ a.s.

Hint. Put the $U_m = \sup_{m \geq n} |X_m - X|$
Then for $m \geq n$

$$E[|X_m - X| | F_m] \leq E[U_n | F_m].$$

$$E\{\limsup_m E[|X_m - X| | F_m]\} \leq E\{E[U_n | F_\infty]\} = E[U_n]$$